

Tim Zhang

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EDUCATION

University of California, Los Angeles (UCLA)

B.S. in Computer Science

Los Angeles, CA

Expected May 2027

East Los Angeles College

Early College Program

Rosemead, CA

Aug. 2020 – May 2024

EXPERIENCE

AI Engineering Intern

Summer 2026

IPMD Inc. – Echo AI & EchoSphere Robot

- Built the retrieval-augmented generation pipeline behind the Echo AI mobile app, orchestrating retrieval with LangChain, detecting facial expressions with OpenFace 3.0 and storing embeddings in a Chroma vector database to turn selfie video into real-time emotional-state insight
- Rebuilt core functionality of the EchoSphere robot's touch engine, classifying raw sensor input into gesture types mapped to distinct emotional response states
- Redesigned the robot's 3D model from reference imagery and embedded four emotion particle animations so expressive states render inside the sphere itself

Computer Lab Consultant

Mar. 2026 – Present

UCLA SEASNet

Los Angeles, CA

- Diagnose and resolve software, hardware and network issues for 100s of engineering students and faculty daily, and run training sessions on lab tooling to reduce repeat tickets
- Contribute to lab-wide system improvements with the SEASNet technical team

OKLAHOMA CITY THUNDER – BASKETBALL DATA SOLUTIONS

Applied AI Engineer Project | *Python, PostgreSQL, pgvector, Ollama, FastAPI, Docker*

Sept. 2026

- Built a fully local RAG question-answering backend over box scores, game details, recaps and injury notes, running Qwen 3.5 2B and nomic-embed-text through Ollama with no hosted inference
- Scoped retrieval with deterministic entity extraction (players, subject/opponent teams, dates, ordinals) over SQL before any vector search, using top-k pgvector hits only as fallback
- Kept arithmetic out of the model: margins, scores and per-player aggregates computed in Python with evidence citations recomputed from selected rows, abstaining when no supporting row exists

Software Engineer Project | *Django, PostgreSQL, Angular, TypeScript, Railway*

Sept. 2026

- Built a Django REST API aggregating possession-level data into per-lineup offensive and defensive efficiency metrics, backed by PostgreSQL
- Built an Angular dashboard for coaching and front-office users and deployed the full stack to Railway

PROJECTS

Website Template Cloner | *TypeScript, Node, Playwright, Next.js*

- Built an agent pipeline capturing a page's rendered DOM, stylesheets and screenshot and emitting a self-contained Next.js template with placeholder copy, reproducing an apple.com rebuild to within 1px over a 12,967px page

ESPORTS.DATA | *Next.js, React, TypeScript, Turso (libSQL), Recharts*

- Built an analytics dashboard over 98,000+ professional matches from 2014–2026, exposing champion presence, synergy, counters and lane matchups through dedicated REST routes against a serverless libSQL database

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL, HTML/CSS | **Human:** English, Mandarin, Cantonese

AI & ML: PyTorch, Transformers, PEFT/LoRA, OpenCV, LangChain, LangGraph, CrewAI, RAG, pgvector, ChromaDB, Ollama, MCP

Frameworks: React, Next.js, Angular, Node.js, Express, Django, FastAPI, Tailwind CSS

Tools: Git, Docker, Playwright, Railway, Vercel, Supabase, Redis, Claude Code

Certifications: AI Fluency, Claude Code 101, Claude Code in Action (Anthropic); CS50P (Harvard, 2024)